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Factors influencing candidate blood donors' deferral, retention and blood safety in Tanzania: A prospective cross-sectional study

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Abstract

Background and Objectives: Several Tanzanian regions fall short of annual whole blood donation targets. Understanding what motivates first-time and repeat donors, and whether certain recruitment and retention strategies may affect blood safety, is essential to improving retention and ensuring a safe blood supply. This study aimed to identify variables associated with pre-donation deferral, donor retention and transfusion-transmitted infection (TTI) prevalence in Tanzanian (candidate) blood donors.

Materials and Methods: Between February and April 2023, 1471 (candidate) donors were surveyed at 43 mobile blood collections across 5 Tanzanian regions to collect data on socio-demographics, motivations, incentives received after donation and recruitment methods used. Multivariable logistic regression modelling was performed on a dataset of 675 participants to identify predictors of pre-donation deferral, retention and TTI rates.

Results: Pre-donation deferral was more likely among females, first-time donors, those recruited via multiple methods and those attending collections at public places (all $p < 0.05$). Donor retention was positively associated with increasing age, having received an incentive and donating at Nyarugusu refugee camp (all $p < 0.05$). Donors mainly motivated by receiving test results had significantly higher TTI rates than those wanting to save lives. TTI rates were lower in those donating at schools/universities and Nyarugusu compared to public places.

Conclusion: This study revealed multiple important predictors of pre-donation deferral, retention and TTI rates in Tanzanian (candidate) blood donors. Further research is needed to identify the most effective donor recruitment and retention strategies and to assess the value of targeting specific populations for a more stable blood supply in Tanzania.

Keywords

blood donation, blood safety, donor motivation, donor recruitment, donor retention, sub-Saharan Africa

Highlights

- Tanzanian students and refugee camp residents seem promising target groups for boosting safe and repeat blood donations. Both display low pre-donation deferral and low transfusion-transmitted infection (TTI) rates, and refugees show high retention rates.
- Low body weight is the leading cause of pre-donation deferral. Deferral is more frequent among females, first-time candidates, people recruited through multiple methods and those presenting at public sites.
- Post-donation incentives seem to improve donor retention, whereas seeking test results drives increased TTI rates.

INTRODUCTION

Blood products are part of the World Health Organization (WHO) Model List of Essential Medicines, and should therefore be available at all times in adequate amounts at affordable prices [1]. Nevertheless, not many countries have achieved universal access to safe and sufficient blood products. The 2018 median annual whole blood donation rate in the WHO African region was 5.4 per 1000 population, with 34 of 47 countries falling short of collecting the recommended 10 donations per 1000 population [2]. Only 17 countries collected at least 90% of donations from voluntary, non-remunerated donors (VNRDs) [2], who are considered the cornerstone of a safe and stable blood supply [3].

Annual reports of the Tanzanian National Blood Transfusion Service (NBTS) indicate that several regions fall short of their yearly donation targets. In 2019, NBTS only reached 82.5% of their set target [4]. Compared to the WHO annual target of 10 per 1000 population, only 53.3% was reached. Furthermore, 15.3% of all blood units collected by NBTS in 2019 had to be discarded, mainly (90.2%) due to a positive transfusion-transmitted infection (TTI) result [4]. Together, these numbers show that Tanzania currently lacks the capacity to fully meet the need for safe and sufficient blood products. Strengthening recruitment and retention of VNRDs might help turn the tide. Accordingly, the Tanzania Red Cross Society (TRCS) supports NBTS to raise awareness on the need for VNRDs, to develop blood donor recruitment strategies and to organize blood collections. TRCS in its turn was supported up until June 2023 by Belgian Red Cross-Flanders (BRC-Fl) for their mandate with various projects.

Strengthening donor recruitment and retention requires understanding what drives Tanzanians towards their first and subsequent donations, and confirming findings of previous research. A 2017 scoping review collating evidence from 35 studies across 16 sub-Saharan African countries identified altruism, donating for family and incentives as key motivators for blood donation [5]. A similar 2025 scoping review of 11 studies published since 2010 showed altruism and access to disease testing as main drivers [6]. Each review included only one Tanzanian study, highlighting altruism [7, 8] and donating for family [7, 8] and friends [7].

It also remains unclear how TTI rates in Tanzania may be affected by certain recruitment and retention strategies, such as the method applied (e.g., face-to-face recruitment, online messaging campaigns) and incentives provided (e.g., gifts, food/drinks). A 2023 systematic review showed that predictors of TTI rates vary widely across Southern African Development Community countries, suggesting that the risks are highly country-specific [9]. Performing country-level evaluations is thus essential to understand Tanzania's unique context.

Therefore, this prospective cross-sectional study aimed to investigate the characteristics of the Tanzanian (candidate) blood donors and to uncover factors predictive of pre-donation deferral, donor retention and TTI rates.

MATERIALS AND METHODS

This manuscript adheres to the Strengthening the Reporting of Observational Studies in Epidemiology checklist for cross-sectional studies [10].

Ethics statement

This study was ethically cleared by the Tanzania National Institute for Medical Research, the Tanzania Commission for Science and Technology and the Social and Societal Ethical Committee of Leuven University (KU Leuven), Belgium (file numbers in Supporting Information 1).

Study setting and participants

Data were collected at mobile blood collections organized by TRCS in five Tanzanian regions (Dar es Salaam, Arusha, Songwe, Kigoma, Iringa) between February and April 2023. Anyone presenting for donation was eligible for participation. All participants provided informed consent in advance and were free to end participation any time without consequences. The anticipated sample size, based on the outcome of TTI prevalence rates, was 1104 (details in Supporting Information 1).

Data collection and management

The standard procedure for mobile blood collections organized by TRCS is described in Supporting Information 1. Study participation consisted of completing a study-specific and piloted paper

TABLE 1 Study sample characteristics ($n = 675$).

Characteristics			Number of participants	Proportion of participants (%)
Mobile blood collections				
Region	Arusha ($n = 12$ blood collections)		201	29.78
	Iringa ($n = 5$)		74	10.96
	Kigoma ($n = 5$)		228	33.78
	Songwe ($n = 8$)		172	25.48
Location	Church		61	9.04
	Nyarugusu refugee camp		46	6.81
	School/university	School	406	60.15
		University	73	10.81
	Public	Field	8	1.19
		Football field	1	0.15
		Market	80	11.85
Socio-demographics				
Gender	Male		465	68.89
	Female		208	30.81
	Not available		2	0.30
Age	<18		54	8.00
	18–24		479	70.96
	25–39		115	17.04
	40–65		27	4.00
Level of education	No education		26	3.85
	Primary school		90	13.33
	Ordinary secondary school		388	57.48
	High school		95	14.07
	University/college		74	10.96
	Not available		2	0.30
Literacy	Literate		644	95.41
	Illiterate		31	4.59
Citizenship	Tanzanian		629	93.19
	Refugees from Burundi		10	1.48
	Refugees from Democratic Republic of Congo		35	5.19
	Not available		1	0.14
Motivations for blood donation				
Type of donor	VNRD		647	95.85
	Replacement donor		9	1.33
	Not available		19	2.81
Main motivation	Altruistic	I want to save the life of another person	586	86.81
		A family member/friend needed blood in the past	10	1.48
		It gives me a good feeling	2	0.30
	Egoistic	I would receive an incentive	0	0
		I receive a medical health check	12	1.78
		I will receive results for blood infections	13	1.93
	Not available		51	7.56

TABLE 1 (Continued)

Characteristics			Number of participants	Proportion of participants (%)
Incentives				
Incentive received after donation (multiple answers possible)	Biscuit		559	82.81
	Soda		547	81.04
	Water		440	65.19
	Rice		117	17.33
	Candy		10	1.48
	Pen		1	0.15
	Cash		1	0.15
	T-shirt		0	0
Would continue donating without incentives	Definitely yes		595	88.15
	Probably yes		56	8.30
	Probably no		6	0.89
	Definitely no		12	1.78
	Not available		6	0.89
Recruitment methods				
Method used to recruit for this blood donation (multiple answers possible)	TRCS volunteer		556	82.37
	Remote	Text message	67	9.93
		Phone	15	2.22
		Social media	19	2.81
	In-person	Family/friend	13	1.93
		Donor Club member	75	11.11
		Other—Church/mosque advertising or leader	17	2.52
		Other—Street/market/field advertising	7	1.04
		Other	9	1.33
Recruited via multiple methods (i.e., >1)			97	14.37
Donation experience				
Overall experience	Good		494	73.19
	Somewhat good		142	21.04
	Neutral		12	1.78
	Somewhat bad		3	0.44
	Bad		16	2.37
	Not available		8	1.19

Note: All data are self-reported.
Abbreviations: TRCS, Tanzania Red Cross Society; VNRD, voluntary non-remunerated donor.

questionnaire in English or Kiswahili (more information in Supporting Information 1, full English versions in Supporting Information 2).

The questionnaire was used to collect self-reported data on (1) socio-demographic variables (gender, age, education level, literacy, citizenship); (2) donor behaviour (number of previous donations, VNRD [gives blood on their own free will without payment] versus replacement donor [friend/family member who donates blood to replace the unit(s) used for a patient's transfusion]); (3) motivations for blood donation; (4) incentives received post donation; (5) recruitment methods and (6) overall donation experience.

The questionnaire also collected data on two of the three outcomes of interest: deferral rates and donor retention rates. Donors were 'deferred' if they indicated they could not donate through failure to meet the NBTS criteria as assessed by the medical personnel evaluating the donor, including weight and age, medical questionnaire responses and a health check. Donors were 'retained' if they indicated that this was not their first blood donation. Data on the third outcome of interest (TTI rates) were obtained after testing by NBTS. The data management process is explained in Supporting Information 1.

Statistical analysis

Donor demographics, motivational and deferral reasons, deferral rates, TTI rates and retention rates were summarized by descriptive statistics using (1) means, standard deviations (SDs) and 95% confidence intervals (CIs) for continuous variables; (2) median and interquartile ranges (IQRs) for non-normal continuous data and (3) percentages for categorical data.

To relate various predictor variables to the three response variables of interest (deferral rates, donor retention rates, TTI prevalence rates), logistic regressions were performed in RStudio: Integrated Development Environment for R version 2024.09.0 [11].

To increase statistical power, categories were grouped as shown in Table 1 for certain variables (i.e., main motivations, recruitment methods, location of collection, region of collection). Recruitment by a TRCS volunteer was chosen as the reference category because this is an important and frequently adopted recruitment method. Kigoma was selected as the reference category for the region of collection, as it accounted for the largest number of records.

Univariable models were used to estimate the odds ratio (OR), 95% CI and *p*-value. Gender, age, as well as variables with a *p*-value of <0.10 in univariable analysis were included in a multivariable model to estimate the adjusted OR (aOR). Statistical significance was set at 5%.

RESULTS

Final study sample

A total of 43 mobile collections across five regions yielded 1471 participant records. However, systematic data entry errors in Dar es Salaam led the BRC-FI researchers to exclude the 784 records (from 13 collections) from this region because of lack of direct access to the TTI data and strong concerns about data reliability. An additional 13 records were removed because they could not be matched with a paper questionnaire, leaving 675 records for analysis, thus falling short of the calculated sample size of 1104. Since it is underpowered, our study is at increased risk of type II statistical error, and the estimates' precision may be limited.

Sample characteristics

Table 1 depicts the characteristics of the final sample of 675 records.

Although only five collections were organized in Kigoma, this region accounted for a third of the records. The majority of participants presented at mobile blood collections organized in schools and universities. One mobile blood collection was organized in Nyarugusu refugee camp in Kigoma on International Women's Day.

The sample was over two-thirds male, with a median [IQR] age of 19 [18–23] years. The vast majority received education at ordinary secondary school or higher level, were literate and were Tanzanian residents. Forty-six participants were refugees from either Burundi or the Democratic Republic of Congo residing in the Nyarugusu refugee camp.

Almost 96% of participants were VNRDs. The most frequent main motivation for the current donation was 'I want to save the life of another person' (87%). Ninety percent indicated they received at least one incentive post donation (e.g., biscuits, water, soda, rice). Only 2.6% of people indicated that, if incentives were no longer provided, they would probably/definitely not continue to donate.

Eighty-two percent were asked by a TRCS volunteer to donate blood. Fourteen percent were recruited through multiple methods. All but one of these 97 participants indicated that this included recruitment by a TRCS volunteer. Overall donation experiences were generally positive.

Donor deferral

Of the 675 participants, 164 were deferred from donating, primarily due to low body weight ($n = 92$), underage ($n = 37$) or low haemoglobin levels ($n = 17$). Other pre-donation deferral reasons are presented in Table S1 in Supporting Information 4.

Logistic regression model analyses results are shown in Table 2. Univariable analysis revealed that gender, the number of previous donations, recruitment method used and location of collection statistically significantly predicted deferral (all $p < 0.05$).

In addition to these variables, participant age and the region of collection were entered into the multivariable model. Gender, the number of previous donations, recruitment method used, location of collection and region of collection were statistically significant predictors of deferral. More specifically, males were statistically significantly less frequently deferred than females (aOR: 0.64, 95% CI [0.43, 0.95], $p = 0.03$), first-time donors were more frequently deferred than repeat donors (aOR: 2.49, 95% CI [1.56, 3.98], $p < 0.001$) and participants at collections organized in schools/universities had lower deferral rates than those attending collections at public places (aOR: 0.28, 95% CI [0.14, 0.54], $p < 0.001$). Despite not reaching statistical significance in the univariable model, region of collection was found to be a statistically significant predictor in the multivariable model. Arusha and Iringa had lower deferral rates compared to Kigoma ($p < 0.01$ and $p < 0.05$, respectively).

To investigate whether the observed effects of location and region of collection on deferral rates might be modified by gender, we ran two additional exploratory bivariable models (see Supporting Information 3). These revealed that the effect of gender on deferral rates was most prominent in public locations and in Songwe.

Donor retention

Of the 675 participants, 285 (42%) indicated that they had donated at least once in the past (see Table S1).

Univariable analysis showed that age, having received an incentive post donation, the location and region of collection statistically

TABLE 2 Logistic regression model for deferral rates ($n = 675$).

Variable		Number of deferred/total participants	Univariable analysis		Multivariable analysis	
			OR [95% CI]	*p*-value	aOR [95% CI]	*p*-value
Age	-	-	0.99 [0.96, 1.01]	0.36	0.97 [0.94, 1.01]	0.11
Gender	Male	98/465	**0.59 [0.41, 0.85]**	**0.005****	**0.64 [0.43, 0.95]**	**0.03***
	Female	65/208	Ref.			
Number of donations	First-time donor	119/381	**2.51 [1.71, 3.69]**	**<0.001*****	**2.49 [1.56, 3.98]**	**<0.001*****
	Repeat donor	45/294	Ref.			
Donor type	VNRD	143/647	0.57 [0.14, 2.30]	0.43	-	
	Replacement donor	3/9	Ref.			
Main motivation	Family/friend needed blood in past	4/10	2.12 [0.59, 7.63]	0.25	-	
	Good feeling	0/2	0 [0, NE]	0.98	-	
	Medical health check	2/12	0.64 [0.14, 2.94]	0.56	-	
	Receive an incentive	0/0	-		-	
	Receive results for blood infections	1/13	0.27 [0.03, 2.06]	0.21	-	
	Other	0/1	0 [0, NE]	0.99	-	
	Save a life	140/586	Ref.			
Main motivation (grouped)	Altruistic	144/598	0.90 [0.53, 1.56]	0.72	-	
	Egoistic	20/77	Ref.			
Recruitment method	In-person	18/55	1.83 [1.00, 3.36]	0.05	1.97 [0.95, 4.11]	0.07
	Remote	15/58	1.31 [0.70, 2.46]	0.40	1.07 [0.55, 2.09]	0.84
	Multiple	33/97	**1.94 [1.21, 3.12]**	**0.006****	**2.30 [1.34, 3.96]**	**0.002****
	TRCS volunteer	97/462	Ref.			
Location of collection	Church	23/61	1.19 [0.60, 2.35]	0.62	1.32 [0.59, 2.96]	0.50
	Refugee camp	7/46	**0.35 [0.14, 0.88]**	**0.03***	0.33 [0.11, 1.01]	0.05
	School/university	104/479	**0.55 [0.33, 0.89]**	**0.02***	**0.28 [0.14, 0.54]**	**<0.001*****
	Public	30/89	Ref.			
Region of collection	Arusha	38/201	0.77 [0.48, 1.23]	0.27	**0.48 [0.28, 0.82]**	**0.007****
	Iringa	18/74	1.06 [0.57, 1.96]	0.85	**0.30 [0.11, 0.85]**	**0.02***
	Songwe	55/172	1.55 [1.00, 2.42]	0.05	0.67 [0.36, 1.26]	0.21
	Kigoma	53/228	Ref.			

Note: Statistically significant results are highlighted in bold.
Abbreviations: aOR, adjusted OR; CI, confidence interval; NE, not estimable; OR, odds ratio; Ref., reference category; VNRD, voluntary non-remunerated donor.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

significantly predicted retention (see Table S2 in Supporting Information 4). All of these variables remained statistically significant in the multivariable model. More likely to be repeat donors were older people (aOR: 1.06, 95% CI [1.02, 1.10], $p < 0.001$), people who received an incentive post donation (aOR: 6.80, 95% CI [2.57, 17.89], $p < 0.001$) and people donating blood at the Nyarugusu refugee camp (compared to public places; aOR: 27.03, 95% CI [2.74, 266.71], $p = 0.005$). Compared to those in Kigoma, people donating in Arusha and Songwe were less likely to be repeat donors (both $p < 0.001$). Despite not reaching statistical significance in the univariable model, the recruitment method was a statistically significant predictor in the multivariable model. Compared to those recruited via a TRCS volunteer, people recruited via multiple methods were less likely to be repeat donors (aOR: 0.57, 95% CI [0.34, 0.95], $p = 0.03$).

TTI prevalence

For these analyses, the sample consisted of the 511 participants who actually donated blood. TTI prevalence rates were 0.44% for human immunodeficiency virus, 1.93% for hepatitis B virus (HBV), 0.89% for hepatitis C virus and 1.33% for syphilis (see Table S1). None of the participants tested positive for multiple TTIs.

Age, the main motivation for donation, having received rice post donation, the location of collection and the region of collection were all statistically significant predictors of TTI prevalence rates in univariable analysis (see Table S3 in Supporting Information 4).

In the multivariable analysis, the main motivation for blood donation, location, and region of collection remained statistically significant. People stating that their main motivation was to receive results

for blood infections had significantly higher TTI positivity rates than those mainly motivated by wanting to save another person's life (aOR: 4.59, 95% CI [1.03, 20.43], $p = 0.045$). TTI rates were lower in participants donating at schools/universities (aOR: 0.07, 95% CI [0.01, 0.46], $p = 0.006$) and the Nyarugusu refugee camp (aOR: 0.04, 95% CI [0.00, 0.34], $p = 0.005$), when compared to participants donating at public places. Arusha, Iringa and Songwe had lower TTI rates compared to Kigoma ($p < 0.01$, $p < 0.05$ and $p < 0.05$, respectively).

Additional analysis revealed that the positive predictive relationship between age and TTI prevalence rates (i.e., the older the donor, the higher the TTI rates) was only present in Arusha and Songwe (see Supporting Information 3). There was no interaction between age and the location of collection.

DISCUSSION

This cross-sectional study collected data on the socio-demographic characteristics, motivations and deferral of 675 (candidate) blood donors presenting at mobile collections across four different Tanzanian regions. We investigated associations between these variables and three outcomes of interest: deferral rates, donor retention rates and TTI prevalence rates.

Five variables were significantly associated with higher deferral: being female, being a first-time donor, attending collections at public places (vs. schools/universities), attending collections in Kigoma (vs. Arusha and Iringa) and being recruited via multiple methods (vs. being recruited by a TRCS volunteer only).

Donor retention (i.e., being a repeat donor) was positively associated with increasing age, having received an incentive post donation and donating blood at the Nyarugusu refugee camp (vs. public places). Compared to those in Kigoma, people donating in Arusha and Songwe were less likely to be repeat donors.

Blood donors mainly motivated by receiving test results for blood infections had significantly higher TTI positivity rates than those indicating that they mainly wanted to save another person's life. Compared to those donating at public places, participants donating at schools/universities and the Nyarugusu refugee camp displayed lower TTI rates. Compared to Kigoma, there were lower TTI prevalence rates in Arusha, Iringa and Songwe.

Several strengths allowed our study to distinguish itself from other cross-sectional studies previously performed in Tanzania. Unlike the previous studies that obtained data by retrospectively accessing donor records [12–15], we surveyed blood donors prospectively on their demographic characteristics and motivations using a questionnaire specifically designed for this study, in order to look for any associations with retention and blood safety. By recruiting participants at the mobile collection sites themselves, we could also survey deferred (candidate) donors to investigate associations with pre-donation deferral.

Moreover, our study included (candidate) donors presenting at mobile blood collections across different regions of Tanzania, whereas previous studies collected data at single fixed blood donation centres in a single region, such as the Northern Zone Blood Transfusion Centre in the Kilimanjaro region [14, 15] or the Muhimbili National Hospital Blood Donation Centre in Dar es Salaam [12, 13].

Despite these strengths, our study has its limitations. Systematic faulty data inputting in Dar es Salaam and lack of access to confidential TTI results led us to remove 784 dataset records. As a result, our study is underpowered and at higher risk of type II statistical errors. As a direct consequence, we had to group certain categories to increase statistical power during data analysis. For example, phone calls, text messages and social media were grouped into 'remote recruitment methods'. As a consequence, we were not able to determine whether these individual methods may differentially impact any of our outcomes (i.e., phone calls vs. text message vs. social media).

Third, there is a risk of social desirability bias, which consists of participants self-reporting socially acceptable answers instead of providing answers that are reflective of their true opinions or thoughts. This might be especially relevant to the questions surveying the participants regarding their (main) motivations for blood donation. Finally, although multivariable logistic regression modelling was performed, there is a risk of residual confounding.

Regardless of its limitations, our findings align with those of previous Tanzanian studies. In a large sample of 14,377 donors at the Tanzanian Northern Zone Blood Transfusion Centre, Valerian and colleagues found that deferral was significantly more likely in females [15]. However, they also found that deferral rates are higher in family replacement donors and increase with age. Moreover, they identified low haemoglobin levels as the leading cause of temporary deferral, whereas low body weight was the main reason in our study sample.

In another cross-sectional survey of 422 students from four Kilimanjaro region universities, 90% of students reported voluntarism as the main reason for ever having donated blood [16]. Similarly, in our study sample, 86% of participants referred to saving another person's life as their main motivation. Although the overall TTI rate of 4.59% in our study was noticeably lower than those reported in studies originating from Dar es Salaam (8.4% [13] to 15.9% [17]) or Northern Tanzania (10.1% [14]), like in these studies, HBV accounted for most of the infections in our study.

Although our study did not include data from Dar es Salaam, it also shows similarities to a recent cross-sectional study in the country's largest city. In Mgasa 2025, 1200 blood units donated at Muhimbili National Hospital Blood Donation Centre were monitored to search for determinants of blood discard [12]. As in our study, the majority of blood donors were young adult males. Most of the units (85%) were collected from family replacement blood donors, which the authors attributed to the fact that blood was collected at a fixed site in a hospital setting. Similarly, the fact that our study surveyed people presenting themselves at mobile blood collections may help explain why 95% of our participants consisted of VNRDs.

Our study uncovered findings that are not yet supported by other research but may provide valuable insights. Participants recruited via multiple methods were deferred more often than those recruited by a TRCS volunteer. Although this requires confirmation in future studies,

one might hypothesize that repeated outreach might pressure people into showing up for blood collection, despite them knowing that they will be deferred from donation. Another interesting finding is that the 46 people donating at the Nyarugusu refugee camp displayed low deferral rates, low TTI rates and high retention rates. Although this sample is small and these findings should therefore be treated with caution, this may result from having the TRCS's Health Information Team in place, a group of refugees who use music, dance and door-to-door social mobilization to share critical health information [18]. People residing at refugee camps may therefore represent an underutilized source of safe blood donations.

Similarly, our study suggests that it may be worthwhile to specifically target students to improve yearly blood donation rates, as they have low deferral and low TTI rates, potentially because of less risk behaviour. Prior research conducted in Southern Tanzania found that medical students presented a high-awareness, positive-attitude and high-intention group to donate blood, and commented that they could be useful to close the gap to the WHO goal of 10 donations per 1000 population [19].

Further research in a larger sample is needed to find out whether it is worthwhile to target certain populations. In addition, future studies are needed to determine preferential strategies for donor recruitment and retention and to shed light on the barriers to achieving a safe blood supply in Tanzania that is based on regular donations by VNRDs. Until then, recruiters should apply the donor recruitment and retention strategies mentioned in the evidence-based blood donor recruitment in Africa guideline and manual (freely available via https://rodekruis.be/recruitment-materials-for-blood-donors-in-Africa-request-form). This manual, developed by BRC-Fl together with an African expert panel, mentions strategies targeted at, culturally appropriate to and feasible to implement in the sub-Saharan African context.

In conclusion, this study identified key predictors of pre-donation deferral, donor retention and the prevalence of TTIs among Tanzanian (candidate) blood donors. Additional research is needed to help determine which recruitment and retention strategies should be favoured, and whether targeted approaches for specific populations would contribute to building a more stable, safe and sustainable blood donor pool in Tanzania.

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J.L., M.J., N.B. and E.D.B. designed the study, N.B. and A.P. managed the project, M.J., N.B. and A.P. acquired and verified the data, J.L. and H.S. cleaned and analysed the data, E.D.B. supervised the research, J.L. drafted the manuscript and M.J., H.S., N.B., A.P., S.D.L., F.F. and E.D.B. reviewed and edited the manuscript.

CONFLICT OF INTEREST STATEMENT

J.L., M.J., H.S., N.B., A.P., S.D.L., F.F. and E.D.B. are employees of Belgian Red Cross-Flanders (BRC-Fl). Its International Cooperation Department receives (external) funding to increase the number of voluntary, non-remunerated blood donors in sub-Saharan Africa through the implementation of blood donor recruitment and retention programmes.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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